

Enterprise Cloud Migration Strategy and Ethical AI Governance Proposal

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1. Introduction and Architectural Options

Global Retail Dynamics (GRD) currently relies on a monolithic on-premise infrastructure that struggles during seasonal

Option A deploys containerized services orchestrated via Amazon Elastic Kubernetes Service (EKS). This architecture

Option B leverages managed serverless functions through AWS Lambda, Amazon EventBridge, and DynamoDB. Ser

2. Ethical AI Framework and Data Governance

As GRD migrates customer transaction and behavioral logs to the cloud, ethical governance is paramount. We propose:

1. Data Sovereignty & GDPR Compliance: Customer records will be partitioned into regional AWS regions (eu-central)
2. Algorithmic Fairness & Bias Auditing: GRD's personalized recommendation engine utilizes collaborative filtering tha
3. Transparent Consumer Consent: Users are provided an intuitive dashboard allowing real-time data export and imm

3. Implementation Roadmap and Risk Management

The proposed cloud migration is organized into a four-phase rollout over an 8-month transition plan:

- Phase 1 (Months 1-2): Foundation setup, VPC peering, and CI/CD security pipelines.
- Phase 2 (Months 3-4): Migration of non-critical analytics and read-only catalog replicas.
- Phase 3 (Months 5-6): Database live replication and dual-write synchronisation.
- Phase 4 (Months 7-8): Cutover of transactional checkout pipelines with canary deployments.

Key risks include network routing failures and data consistency divergence during dual-write phases. We mitigate these

4. Conclusion and References

The containerized Kubernetes approach represents the most resilient architecture for GRD's high-throughput requirements.

References:

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